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Technical Letter

HIGH TEMPERATURE ELECTRIC CONDUCTIVITY OF PURE SILICON

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Abstract — The electric conductivity of high purity silicon has been measured at elevated temperatures. Thin silicon rods were heated in a horizontal electric resistance furnace up to temperatures of 1120 °C. By passing a current through the rods and by measuring the resulting voltage drop, the conductivity was determined in the temperature range from 550 to 1120 °C. The data compare very well with reported literature values.

Résumé — On a mesuré la conductivité électrique à des températures élevées du silicium de haute pureté. On a chauffé des tiges minces de silicium dans un four électrique horizontal à résistances jusqu'à des températures de 1120 °C. En passant un courant à travers les tiges et en mesurant la chute de potentiel qui en résultait, on a déterminé la conductivité dans la gamme de température de 550 à 1120 °C. Les données se comparent très favorablement aux valeurs rapportées dans la littérature.

INTRODUCTION

The conductivity of silicon rods with electronic grade purity was measured at high temperatures for the purpose of using silicon rods as hot filaments for chemical vapour deposition of solar grade silicon from $SiHCl_3$ gas. In the Siemen's reactor [1] silicon rods are heated to about 1100 °C by the passage of current through them. This temperature is required to promote the decomposition of the tri-chlorosilane into pure silicon. Since all the energy is supplied by the passage of current through the hot silicon rods, it is important to know how the electric conductivity varies with temperature. Using these values it is possible to indirectly estimate the thickness of the rods as they grow within the reactor, which then can be used for process control. The conductivity data are also important for the design of the power supply to be used. As the rods grow in thickness, there will be greater temperature gradients within the rods leading to non-uniform current distribution.

EXPERIMENTAL

A schematic of the experimental setup is shown in Figure 1. A small resistive heating furnace was used to heat a ceramic mullite tube with an inner diameter of 4.3 cm and length of 50 cm. Two thermocouples were used to measure the temperature at two locations to ensure temperature uniformity. The silicon rod was connected to graphite connectors on either end and fastened with graphite screws. The graphite connector was further attached to a 6 mm diameter copper rod allowing for

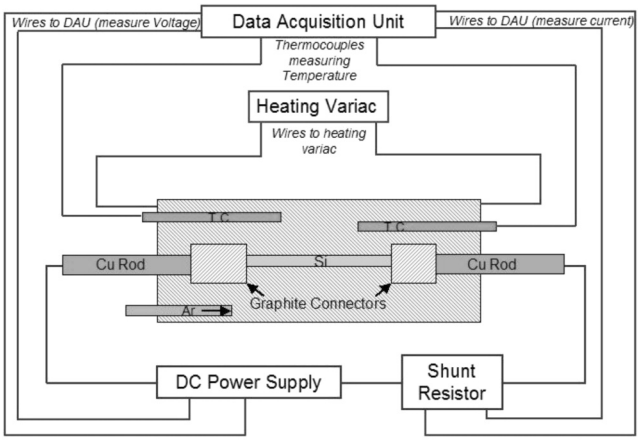


Fig. 1. Experimental schematic for conductivity measurements.

conduction of the electric current. The effective length of the silicon rods was set equal to the distance between the centre of the two graphite screws. The screws pressed firmly on to the silicon rod creating good electric contact between the graphite connector and the silicon rods. The experiments were performed under an argon atmosphere to prevent the oxidation of the silicon and the graphite connectors.

A DC power supply was used to provide a known current through the silicon rod. The current was monitored through a shunt resistor (10A/50mV) and the voltage was read directly from the DC power supply. The current, voltage and corresponding temperatures were logged into a data acquisition unit connected to a computer, to determine the conductivity-temperature relationship.

The silicon rod went through a heating and cooling cycle for each test; therefore there were at least two sets of conductivity data obtained from each test. The furnace was pre-heated to approximately 600 °C before passing current through the silicon rod. The heating of the silicon rod was achieved by the furnace heating and self-heating was minimized by passing a low current ($I \leq 5A$) through the rods during the measurements. In some initial tests with higher rod currents it was found that the silicon rod self heated and it was difficult to measure the actual rod temperature. The temperature range tested was between 550 °C to 1120 °C.

The silicon rod dimensions were carefully measured for each test. The dimensions of the rods used in our tests are shown in Table I. The chemical assays(Glow Discharge Mass Spectroscopy) of the silicon rods show very low level of impurities(0.016 ppm B, 0.048 ppm Fe, 0.018 ppm Cu, 0.33 ppm Cl, 0.021 ppm K). All other elements including P, S, Na, Mg, Al, Mn and Cr reported as less than 0.01 ppm or below detection limit.

RESULTS AND DISCUSSIONS

Based on the current passed through the silicon rod and the voltage drop across the rod, the electric conductivity was determined. The resulting conductivity obtained during the heating and cooling cycles from test T1 is shown in Figure 2. It is observed that there is good correspondence between the data obtained during the heating and cooling cycles. This supports that the temperature of the silicon rod indeed is equal to the temperatures measured using the two thermocouples. During most tests, the voltage applied across the silicon rod was varied depending on the actual temperature and the actual applied power to the silicon rods varied from about 1.0 to 20.0 watts. If this power had increased the silicon rod temperature noticeably above the reactor temperature, we should have noticed differences in the results with the varied powers.

Table 1 – Silicon rod dimensions for conductivity tests

Test Number	Length (cm)	Diameter (mm)
T1	11.3	4.81
T2	11.3	4.81
T3	11.3	4.84
T4	11.3	4.84
T5	5.6	4.90
T6	5.6	4.90
T7	20.3	4.84
T8	20.3	4.84

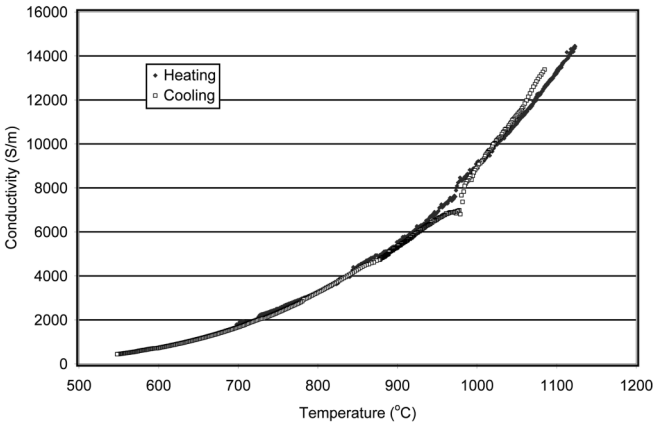


Fig. 2. Conductivity of silicon versus temperature for run T1.

To illustrate the reproducibility between the various rods used, conductivity data are given in Figure 3 for rods with different lengths. From this figure, it is evident that the conductivity values are similar indicating that both contact resistances or temperature variations have negligible effects on the electric conductivity of silicon.

To determine the temperature dependency of the conductivity data, they were fitted to the Arrhenius equation.

$$\sigma = A\,exp\left(-\frac{E}{RT}\right) \tag{1}$$

where σ is the electric conductivity (S/m), E is the activation energy (J/mole), R = 8.314 J/mol·K, T is the absolute temperature (K) and A is a constant (S/m). By taking the logarithm of each side, a simple linear equation can be obtained and the data of each experiment will be fitted to this form:

$$\ln(\sigma) = \ln(A) + \left(-\frac{E}{R}\right)\left(\frac{1}{T}\right) \tag{2}$$

By plotting $\ln(\sigma)$ versus $1/T$, the slope obtained is $(-E/R)$ and

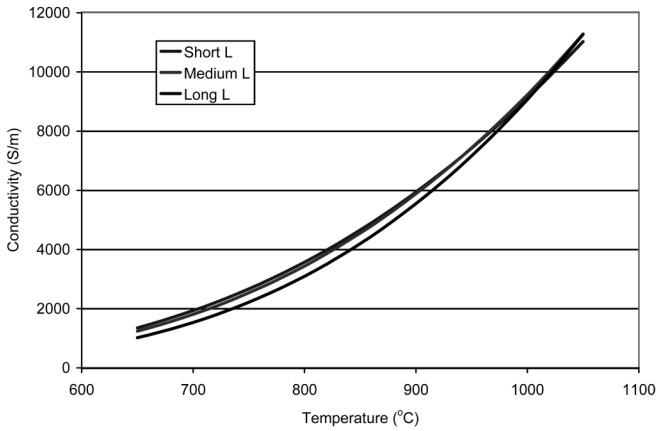


Fig. 3. Conductivity comparison between silicon rods of various lengths.

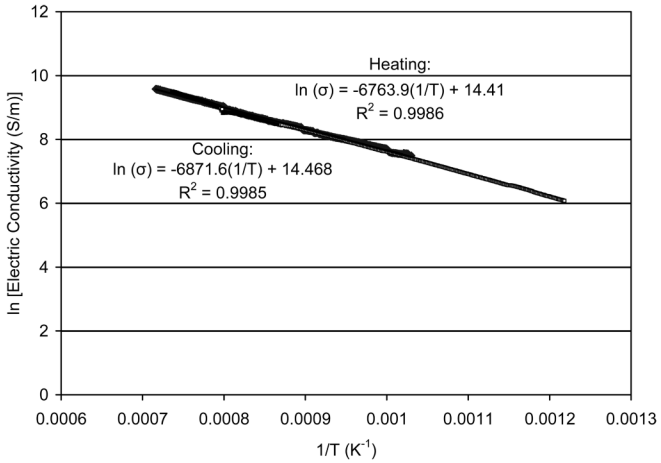


Fig. 4a. Natural logarithm of conductivity versus inverse temperature.

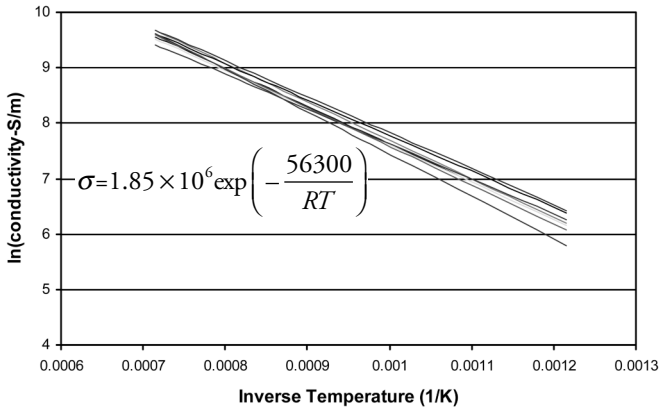


Fig. 4b. Natural logarithm of conductivity versus inverse temperature for all the tests.

the intercept is ln(A), whereby all constants in the Arrhenius equation can be found. A sample of the fitted data is shown in Figure 4a for test # T1. It is seen that indeed a linear relation exists and therefore the Arrhenius equation represents the data well. Figure 4b gives an Arrhenius plot for all the tests carried out. It is seen that there is good agreement between the various rods tested. By regression analysis of all the conductivity data, the following equation was obtained

$$\sigma = 1.85 \times 10^6 \ \exp\left(-\frac{56300}{RT}\right) \ \text{(S / m)} \tag{3}$$

The activation energy (E) is therefore 56,300 J/mole.

The theoretical activation energy can be estimated from the bandgap energy (E_g) of pure silicon. Since the silicon is pure and without any dopant, the conductivity can be associated to the intrinsic conduction which has a reported energy (E_g) of 1.10eV [2]. The intrinsic conduction dependence is shown in the following equation which is derived from theoretical studies [2], where e is the electron charge and μ is the electron mobility.

$$\sigma \propto T^{3/2} \exp\left(\frac{-E_g}{2kT}\right) e\mu \tag{4}$$

By substituting for the Boltzmann constant (k) and rearranging, we find that the activation energy (E) is related to the bandgap (E_g) according to:

$$\begin{aligned} E &= \frac{E_g N_A}{2} = \frac{E_g}{2} \times 1.6022 \times 10^{-19}\ {}^{\text{J}}/_{\text{eV}} \times N_A \\ &= \frac{1.10\ {}^{\text{eV}}/_{\text{e-}}}{2} \times 1.6022 \times 10^{-19}\ {}^{\text{J}}/_{\text{eV}} \times 6.023 \times 10^{23}\ {}^{\text{e-}}/_{\text{mol}} \\ &= 53100\ {}^{\text{J}}/_{\text{mol}} \end{aligned} \tag{5}$$

With this approach, the theoretical activation energy for intrinsic conduction of silicon becomes 53,100 J/mol which is fairly close to the measured value of 56,300 J/mol.

Furthermore, to test the accuracy of our measurements, we compared our results with data previously published [3-8]. Such comparison is shown in Figure 5. Except for the data by Abdel–Aal *et al.* [8], our data is in excellent agreement with values already reported.

Every single set of data shows that as the temperature increases, so does the electric conductivity of silicon. Moreover, at the melting point, the electrical conductivity experiences a sudden increase of about two orders of magnitude. Although it is expected that the properties of any material would be altered during the solid to liquid transformation, this sudden steep increase is dramatic. Since the electric conductivity increases drastically with temperature, it is expected that the electric current flowing through the silicon will not flow uniformly due to uneven

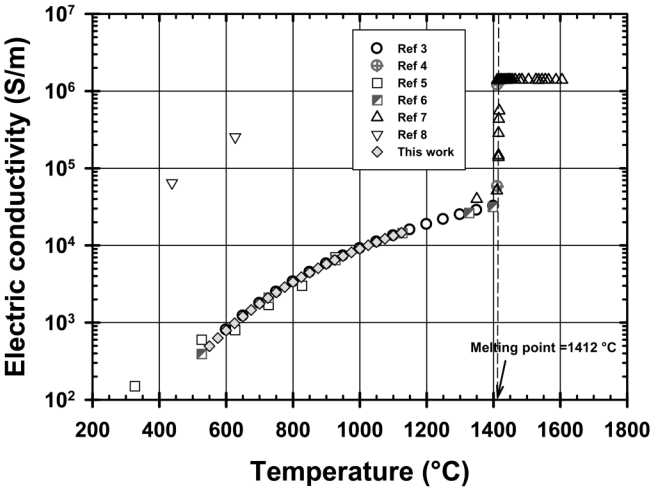


Fig. 5. Comparison of our measured electrical conductivity with data already published.

temperatures within the silicon rods, with their centre hotter than the outside.

CONCLUSIONS

The electric conductivity of pure silicon rods was measured at temperatures between 550 and 1150 °C. The conductivity was found to follow the Arrhenius equation with an activation energy of 56.3 kJ/mol. The results from this investigation compare very well with previously reported data. The measurements were carried out on silicon rods with diameters of about 4.8 mm and with lengths from 56 to 203 mm. The rods were heated in an electric furnace under argon gas and the conductivity was measured by passing a small current through the rods and measuring the resulting voltage drop.

ACKNOWLEDGEMENT

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